

yzcodehub

Graphics Systems Engineer

PUBLIC_EMAIL <https://github.com/yzcodehub>

PROFILE

Graphics systems engineer focused on real-time rendering, engine architecture, GPU compute, cross-platform graphics interfaces, and multimedia systems, while exploring neural graphics through reproducible experiments.

CAPABILITIES

Graphics APIs	Vulkan / DirectX 11/12 / Metal / OpenGL / WebGPU
Rendering	PBR / IBL / Shadow / SSAO / Post-processing / Render Graph
Systems	Modern C++ / RHI / ECS / GPU Compute / Resource Lifetime / Profiling
Media & Learning	FFmpeg / A/V Sync / Zero-copy / Python / PyTorch / ONNX

EXPERIENCE

Professional experience (anonymous)

Graphics and engine systems development

- Designed maintainable runtime boundaries around cross-platform RHI, resource binding, pipeline state, and GPU synchronization.
- Worked on rendering pipelines, geometry processing, and performance analysis using frame time, bandwidth, and resource lifetime.
- Handled data movement, thread coordination, and A/V synchronization in high-resolution multimedia systems.

C++ / Vulkan / DirectX / Metal / GPU Profiling / Multimedia

Ongoing

Independent graphics lab

- Turned low-level graphics concepts into runnable, fallback-safe public demos with Three.js, raw WebGPU, and WGSL.
- Studied the path from graphics-data generation to training and browser inference with PyTorch and ONNX Runtime Web.

TypeScript / Three.js / WebGPU / WGSL / PyTorch / ONNX

PUBLIC LAB

- Render graphs and resource lifetime
- GPU compute and progressive path tracing
- Browser neural-denoising deployment